

E-Commerce Sales & Profit Analytics Dashboard

Power BI portfolio project report

Prepared by Abdul Wahab

Dataset period: January 2023 - December 2025

A three-page business intelligence solution covering executive KPIs, product profitability, customer value, and regional performance.

1. Project overview

This project demonstrates an end-to-end Power BI workflow for an online retailer. It converts transaction-level order data into an interactive dashboard that helps decision makers monitor revenue, profit, customers, products, and geographic performance.

Business questions

- How much sales and profit did the business generate?
- What is the overall profit margin and average order value?
- Which categories and products lead revenue?
- Which products generate losses?
- How do monthly sales change over time?
- Which regions, cities, and customers perform best?

Dataset profile

Order lines	Orders	Customers	Period
1,953	1,150	239	2023-2025

The dataset contains order identifiers, dates, shipping mode, customer and segment fields, region, state, city, category, sub-category, product, quantity, discount, sales, profit, and shipping cost. It is synthetic and designed for portfolio use.

Tools and techniques

Power BI Desktop, Power Query, DAX, date modeling, KPI cards, Top N filtering, conditional formatting, slicers, cross-filtering, bookmarks, and a custom report theme.

2. Data preparation and model

Power Query preparation

The CSV is imported with UTF-8 encoding. Headers are promoted, dates are converted to Date, quantity to Whole Number, discount to Percentage, and sales, profit, and shipping cost to Currency. Blank orders are removed and Row ID is used for duplicate checks.

Data model

A dedicated Date table supports chronological sorting and year-over-year analysis. The model uses a one-to-many relationship from Date[Date] to Sales[Order Date]. Measures are stored in a separate Measures table for easier maintenance.

Core measures

```
Total Sales = SUM ( Sales[Sales] )

Total Profit = SUM ( Sales[Profit] )

Total Orders = DISTINCTCOUNT ( Sales[Order ID] )

Total Customers = DISTINCTCOUNT ( Sales[Customer ID] )

Profit Margin = DIVIDE ( [Total Profit], [Total Sales], 0 )

Average Order Value = DIVIDE ( [Total Sales], [Total Orders], 0 )
```

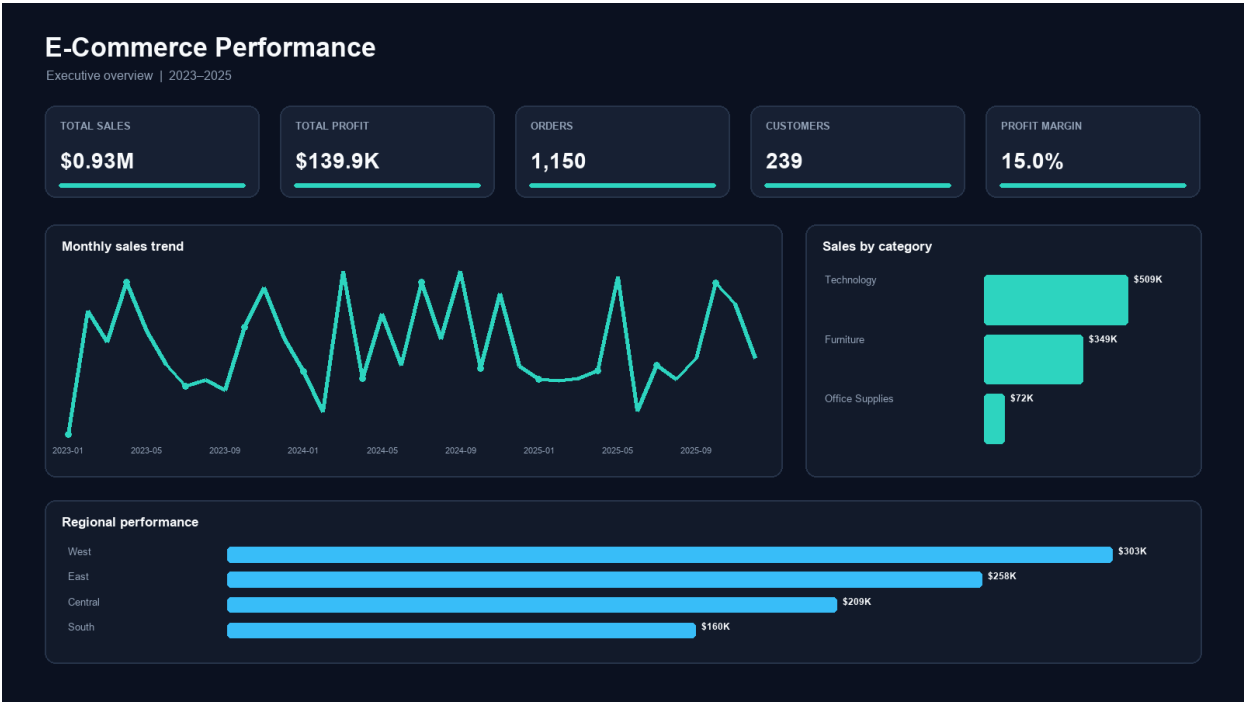
Additional measures include prior-year sales and profit, year-over-year growth, product and customer rankings, total quantity, average discount, and loss-making product count.

Quality checks

The dashboard baseline should reconcile to \$930,546.35 in sales, \$139,905.81 in profit, 1,150 distinct orders, 239 distinct customers, a 15.03% profit margin, and an \$809.17 average order value when no slicers are applied.

3. Executive Overview

Page 1 provides the headline KPIs, monthly sales trend, category contribution, and region performance. Date, region, and segment slicers allow executives to move from the overall result to a focused view.

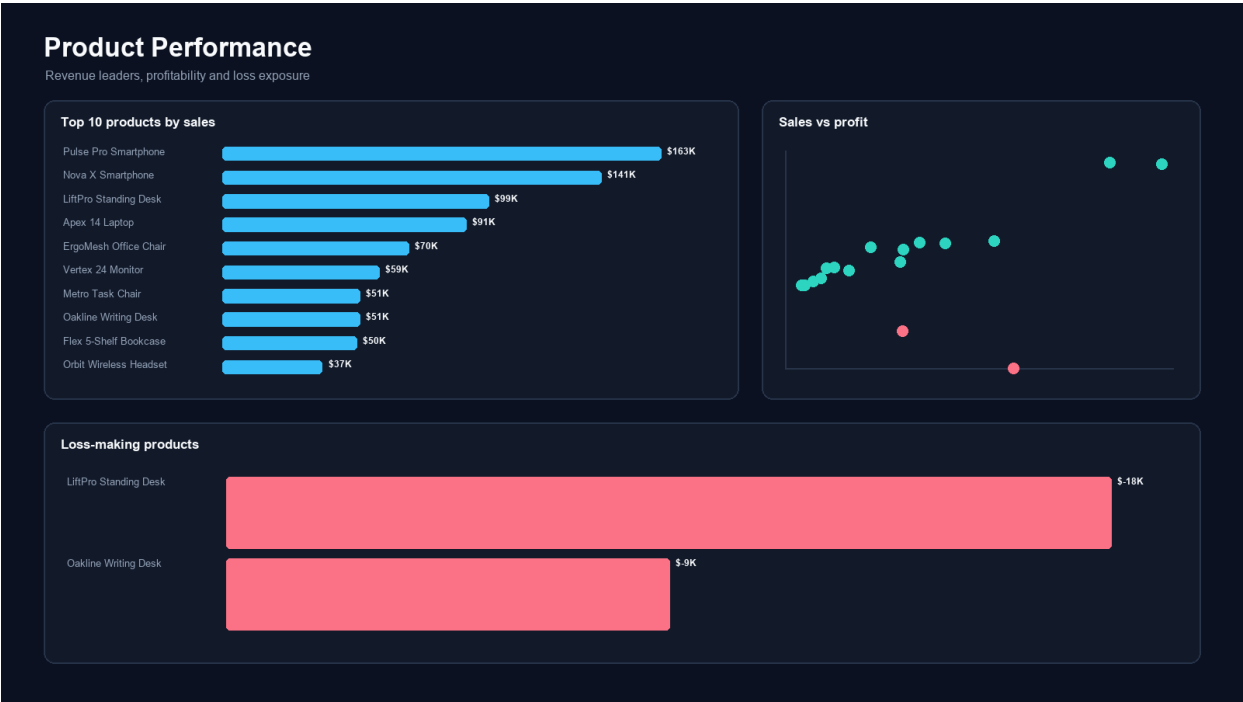


Key visual elements

Five KPI cards, a monthly line chart, category bars, regional comparison, tooltips, and synchronized slicers.

4. Product Analysis

Page 2 identifies the top 10 products, compares sales with profit, and isolates loss-making items. The scatter plot makes high-sales/low-profit risks visible, while the loss table supports corrective pricing and discount decisions.

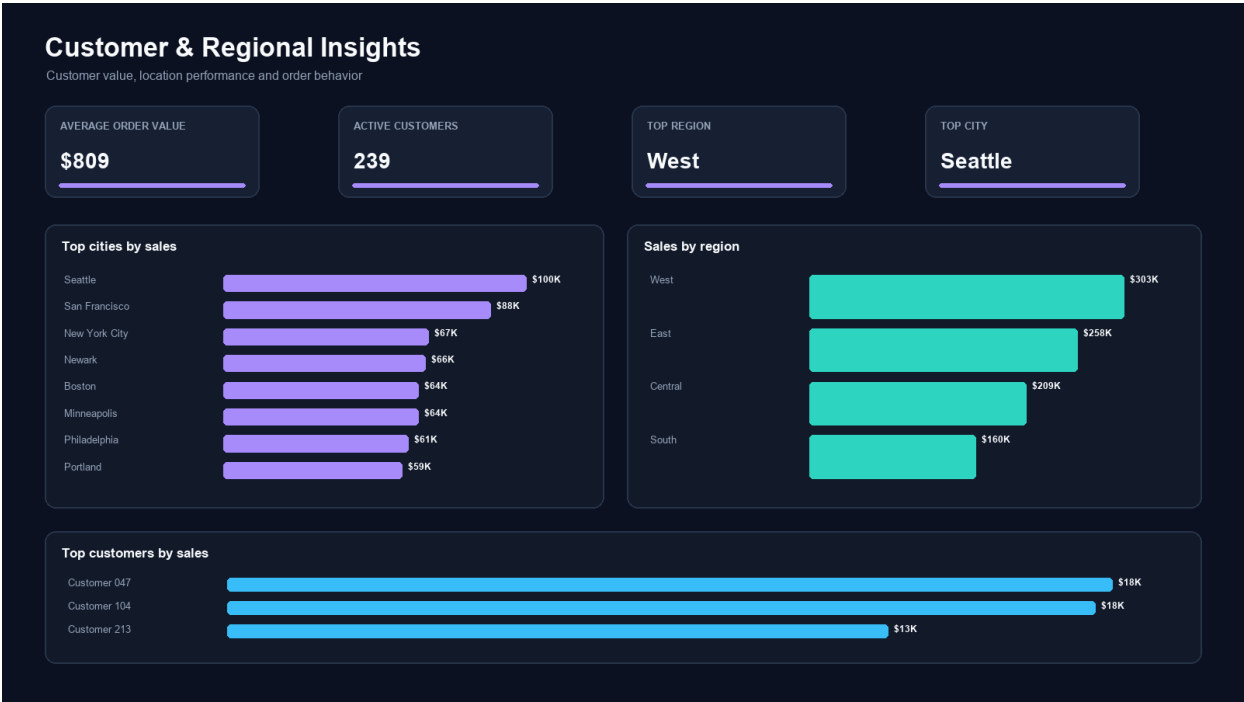


Key visual elements

Top-product ranking, category and sub-category matrix, sales-versus-profit scatter plot, and a loss table filtered to negative profit.

5. Customer and Regional Analysis

Page 3 compares cities and regions, surfaces the highest-value customers, and monitors average order value. Region, state, segment, ship mode, and date slicers support detailed customer and location analysis.



Key visual elements

Average order value, customer count, city ranking, regional comparison, customer table, and optional map visual.

6. Findings and recommendations

Key findings

Sales	\$930,546 total sales across 1,150 orders.
Profitability	\$139,906 profit and a 15.0% overall margin.
Category leader	Technology generated the highest sales.
Regional leader	West was the strongest region, while Seattle led cities.
Product leader	Pulse Pro Smartphone generated the highest product sales.
Loss exposure	LiftPro Standing Desk lost about \$18.2K and Oakline Writing Desk lost about \$9.1K.

Recommended actions

- Review the cost, shipping expense, and discount policy for the two loss-making desks.
- Protect Technology availability because it is the largest sales contributor.
- Replicate successful West-region tactics in weaker regions while checking whether customer mix explains the gap.
- Track year-over-year growth and margin together so revenue gains do not hide profitability erosion.
- Use customer ranking to support retention campaigns for high-value customers.

Repository contents

The repository includes the cleaned CSV dataset, DAX measures, Date table, Power Query script, dark theme JSON, three dashboard previews, build instructions, README, validation summary, this report, and an MIT License.

Portfolio description

Developed an interactive Power BI dashboard to analyze sales, profitability, product performance, customer behavior, and regional trends. Used Power Query for data transformation, a date model for time analysis, and DAX for KPI calculations, rankings, and loss analysis.

Prepared for the Power BI Data Visualization portfolio of Abdul Wahab.